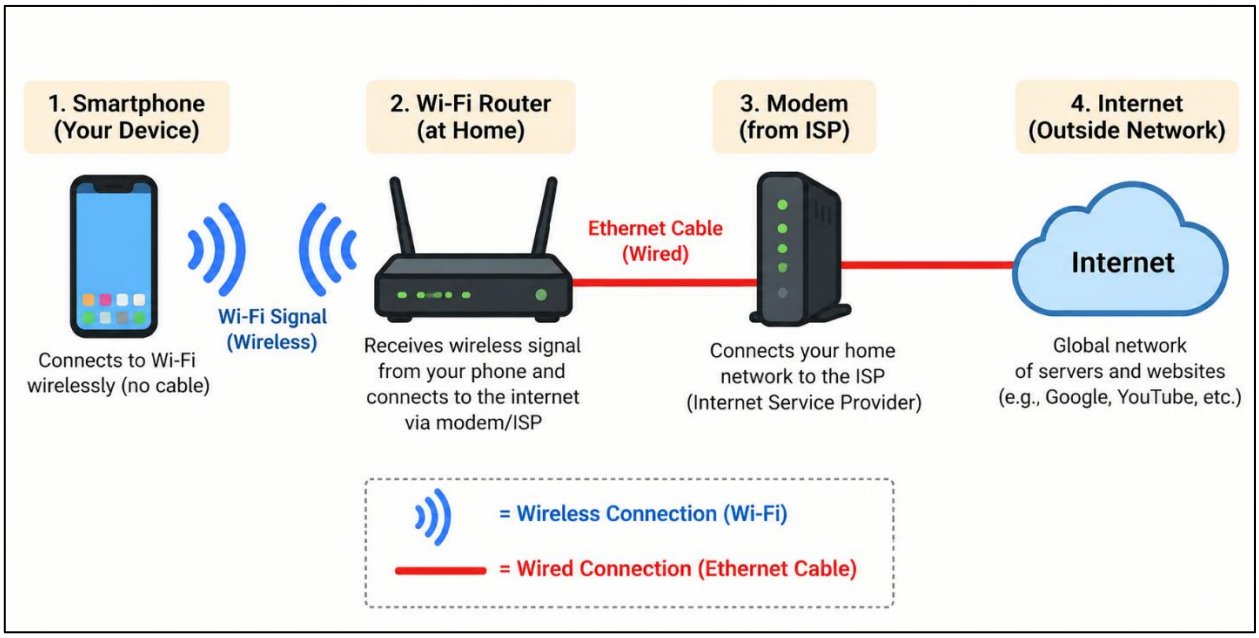
1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standard	Frequency Band(s)	Maximum Theoretical Speed
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps
802.11ac (Wi-Fi 5)	5 GHz only	Up to 3.5 Gbps
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Up to 9.6 Gbps

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

Available Wi-Fi Networks (Example)

No.	SSID (Wi-Fi Name)	Signal Strength	Security Type
1	Home_WiFi	Excellent (95%)	WPA3
2	Airtel_5G	Good (75%)	WPA2
3	Public_WiFi	Weak (40%)	Open (No Security)

I would choose **Home_WiFi** because:

1. **Strong Signal:** It has 95% signal strength, providing a fast and stable connection.

2. **Better Security:** WPA3 offers stronger protection than WPA2.
3. **Safe Connection:** It is more secure than an open public Wi-Fi network.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID hiding is a setting on a router that stops it from broadcasting its Wi-Fi name to everyone nearby.

Normally, a router constantly shouts its name into the air so your phone or laptop can see it in the Wi-Fi list. When you hide the SSID, your network name becomes "invisible." To connect a new device, you cannot just click it—you have to manually type in the exact name and password.

One Advantage: Hiding from Neighbours

The main benefit is **staying out of sight**. Because your network name does not show up on standard Wi-Fi lists, casual snoops, neighbours, or people walking past your house will not even know your network exists. They are much more likely to try to connect to or mess with a network they can actually see.

One Disadvantage: False Safety and Battery Drain

The biggest problem is that **it does not actually make you safe, and it can drain your battery**.

Anyone with free hacking software can still see your network's hidden traffic in seconds. Even worse, because the router is now quiet, your phone or laptop must constantly shout into the air, *"Are you there, Home Network?"* wherever you go. This uses more battery and tells strangers out in public the name of your private home Wi-Fi.